

Section 3 - Building a Garment - Sewing



Lecture 4 | Avatars

Hello everyone, this lesson will cover the basics on Avatars and Arranging your garment. You can follow along with your own file or by downloading the practice file.

AVATARS

- Library→ Avatars
- Double Click or Right Click to add to Workspace.
- Version 6.2 - 16 humans, and 5 geometric shapes
 - 6 male, 6 female and 4 kids
 - Many Styling options
 - hair, poses, shoes, motion files (which is animation), texture (which is their skin).
 - Easy to add; double click in Library, Right click to remove
- Default Avatars have joints Rigged with a skeleton
 - Move joints & Save as a pose.
- Default Avatars are editable by size.
 - The Avatar editor allows you to change their points of measure.
 - basing their Avatar off their dress form and sometimes you may only have a few basic points of measure, so we have an option for that as well.
- CLO Also now includes ASTM Sizing.
 - Available in the Library.
 - Eg. Female Avatars have 63 different ASTM presets you can apply, including Maternity.
- Change hair, shoes, poses, etc.; create your own without a lot of hassle.
- Adult Avatars - Body Styling
 - Easy appearance adjustment
- Child Avatars
 - Texture Folder → Drag and drop the texture into place.

ARRANGEMENT

- Practice File
- Arrange the wearer's right side to look like the wearer's left.
- Best Practice Tips
 - 3D Toggle Menu→ Textured Surface; front & back of Fabric
 - Move my pattern out of the way; this will help me to see the arrangement points.
 - 2 key to bring myself front and center.
 - Use your 2D window for selecting things.
 - not trying to drag pattern pieces towards your Avatar, which you do not need to do when you're arranging.
 - The way to arrange is to use these "blue dots" or arrangement points.

- o 3D Vertical toggle menu→ Avatar Display→ 2nd icon arrangement points.
 - o "blue dots" on Avatar. (Shortcut; Shift +F)
- Select a pattern piece
 - o wearer's right side in the 2D window.
 - o click once, then as I hover over the blue dots; ghost outline of pattern
 - o NOT dragging pattern pieces into place,
 - o Pattern where you want; click that blue dot,
- The pattern piece sits between the blue dots and your Avatar.
- The pattern piece is highlighted yellow and curved around the Avatar
- The Arrangement Point has turned from blue to yellow.
- Note; sleeve and my cuff are arranged in the 2D window→ 3D they are also sitting nicely together.
 - o Arrange patterns together
- Exact placement is not important.
 - o CLO will sort this out when sewn/simulated
 - o After we sew we can move pattern pieces to be sure important areas wrap around limbs
- Leggings front→ Marquee select.
- Place them together.
 - o Lay out nicely in 2D window for easy placement.
 - o Ok to do them individually, better together.
- Note CLO places back of fabric towards Avatar→ charcoal surface.
 - o Thick Textured Surface; both sides and thickness.

Lecture 5 | The Gizmo

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- Specific Practice File
 - Arrangement points here, it's going to wrap the skirt
 - o Property Editor→ Offset; better
 - o Property Editor→ Flat; OK
 - Undo and turn arrangement points off.
 - Select in either window
 - All 3D applications have Gizmos; allow you to move these pattern pieces in 3D window.
 - Mentioned previously; Not super important that things are perfectly arranged. Note following;
 - o On the correct side of the Avatar,
 - o Arranged in the correct places next to each other.
 - o Fully wrap around the arm/leg/neck so the seams come together.
 - o Do not have major sewing lines intersecting the body.
 - Note threads through arms
 - Haven't learned simulation yet.
 - Simulate to see errors. Stop & Undo CMD or CTRL-Z
 - o Simulation button or Spacebar
 - Use Gizmo to move/rotate Patterns
 - o Near body/avoid arms
 - o Gizmo [default Screen Coordinate Gizmo]
 - o DIFFERENT GIZMOS
 - No better or worse gizmo; personal preference.

- Right-Click 3D window background to change
- Screen Coordinate - based on the camera axis. Change the camera view in order to change the gizmo axis. Selected object moves along with the camera axis
 - "Your point of view"
- Local Coordinate - based on following the native direction of an object. Mainly usable when designing poses from the avatar's X-ray joint points. Selected object moves to native direction of an object.
 - "Other 3D programs; X, Y and Z axis constant, forward and backward always going to be forward and backward
 - useful if you want things to always be at right angles and perpendicular or square
- World Coordinate - based on 3D space regardless of screens or objects. Selected object moves based on 3D space regardless of screens or objects
- Central cube
 - faces move along planes
- 3 AXIS'
 - Green→ Y Axis "up & down"
 - Red→ X Axis "left to right"
 - Blue→ Z Axis "front to back"
 - circles rotate around the corresponding Axis
 - box moves along that plane
- Gizmo Target
 - Click the target then you click on Avatar
 - Place the pattern piece on that spot;
 - Not curved
 - Quick pattern piece placement
 - Good for small patterns [pockets, etc]

Lecture 6 | Reset Arrangement

Practice File

- Arrangement points register the location of the pattern within the software. R
- Remember where you put it; easy to go reset
- Reset 3D Arrangement Tool, here in the 3D Toolbar.
 - All patterns will snap to the arrangement
- Initial simulation is bad - Reset 3D arrangement fix position. Simulate again

Reset Individual Pieces

- Example; Right Sleeve & Right Hood
- Select Pieces. Right Click in 3D Window→ Reset 3D Arrangement Selected.

Reset 2D Arrangement

- Reset 2D Arrangement Tool→ reset everything to arrangement in 2D window.
 - Negate your 3D arrangement

- Be sure Patterns arranged well in 2D window

Reset Individual Pieces

- Reset one piece to get it to be flat again.
- Select Pieces. Right Click in 3D Window→ Reset 2D Arrangement Selected

Hello everyone. In this tutorial video we will begin sewing in CLO. I invite you to follow by using your own files or by downloading the exercise files from the CLOSET link in the description of the video below. Let's get started.

Lecture 7 | Segment Sewing

- Location in toolbars - 2D & 3D
- Most common sewing tools - segment & free sewing
- Start with Segment Sewing
 - Sew together 2 full Segments
 - Important that lengths match.
 - Excess will be distributed evenly and will appear smocked or puckered.
 - Not recommended if you have ease that needs to be placed.
 - To use this tool, click twice; simply left click once on each Segment
 - Note as I hover; it highlights blue
 - Note after click - marks designate the start of the sewing,
 - Place marks on the segments in the same direction.
 - Click to sew segments together.
 - View in 3D with your sewing lines;
 - If Twisted; reverse sewing CMD/CTRL-B immediately
 - Edit with Edit Sewing Tool
- Symmetric Garment
 - Sew using Segment Sewing
 - Sewing applied to both sides at the same time
 - Check in 3D window for errors
 - Sew center front & back
- Segment Sewing in 3D window; same as 2D window.
 - Note mark at the start
 - 2 clicks

Lecture 8 | Free Sewing

- Pattern has extra segment points; cannot use Segment Sewing
- Click to start, hover and move. click to end
 - CLO will lock you into the direction you take
 - CMD/CTRL Z or ESC to end if necessary

- Note Mark at start of the sewing
 - view in 3D window
- Click on matching segment, starting in same direction
 - Blue picking point appears; stop point
 - appears with or without a segment point
 - Wrong direction; delete or backspace to start matching segment over
 - First segment stays
- Hover and move. click to end.
- 4 clicks.
- If Twisted; reverse sewing CMD/CTRL-B immediately
- Edit with Edit Sewing Tool
- Free Sewing in 3D Window
 - Same as 2D Window BUT double click at each end
 - directions on cursor
- Sweatshirt pocket
 - Sew Pattern to Outline & Pattern to Internal Lines separately

Lecture 9 | Precision Sewing

- Click, hover & move. Right click and precision dialogue will appear
 - Mac Magic Mouse; CONTROL instead of Right Click
 - enter specific length.
 - *PRECISION BOX TECHNIQUE SAME WITH OTHER TOOLS
 - Click OK
- Free Sew other segment; use blue picking point or Precision.

EDIT SEWING

- Sewing machine with the cursor arrow.
- All EDIT tools in CLO; indicated with the little cursor arrow.
- Select and delete it backspace or delete key
- Right-Click
 - to delete
 - reverse the sewing if crossed/twisted. hot key for this is CMD/CTRL B.
 - Select the SeamLine Pair Group & CMD/CTRL B to reverse sewing.
 - Last Sewing is active; CMD/CTRL B to reverse sewing.
- Select SeamLine Pair Group and drag to move it around.
 - select sewing in the middle, it will slide along pattern edge.
- Select only point at either end, move point and change the sewing length.
 - Select poin on other half to move the other side to match,
 - Note blue picking point to stop to match the sewing length.
 - Overlapping or Sewing Ends that match up will show a drop down
 - This happens when clicking on 2 things at once
 - Which one do you want to select?
 - Selected will fully highlight SeamLine Pair Group

- Sewing end points will also snap to each other
 - easy to match without overlapping them.
- Other Edit Sewing Options
 - *similar options as main selection tools, but only apply the sewing.
 - Adding a point at the start or end of the sewing; mark the pattern & add a point.
 - Merge that seam. This is an option also available with the Edit Pattern tool and I find most users access it there more often.
 - Deactivate the sewing. The sewing will turn white
 - temporary technique used for things like twisting part of the garment.
 - *List & review other options
 - CMD/CTRL A to select all sewing

Lecture 10 | 1:M & M:N Sewing

1:M SEGMENT (One to Multiple).

- 1 to M and M to N Sewing File.
- Segment and Free Sewing option to sew 1 seam to multiple seams; 1 to M [Multiple] sewing
- START One total Segment that sews to multiple smaller Segments.
- Sew 1 Segment first before sewing the multiples to it.
- Pay attention to Start mark.
- Will dictate both the direction to sew each pattern, and the order to sew
- Sew the first pattern, press & hold SHIFT, sew the multiple patterns
- Do not release SHIFT until you are finished with all the sewing.
- Pay attention to both the order in which you sew your multiple pieces, as well as which end the ticking mark is on.
- Fail to press shift before the second pattern, first pattern will simply sew to your second pattern.
- Release shift before you finish, Sewing Relationships will be established with the pieces you had sewn before you release shift.

1:M FREE SEWING

- Very similar but used when pieces are segmented poorly.
- Sleeve cap to multiple armhole pieces.
- Sew the 1 piece first. Click to start and click to end on the sleeve cap.
- Press & Hold SHIFT until completing the entire armhole.
- Mess up one step in the process; press delete or backspace on your keyboard and start that step over.
- One section reversed?
 - Different color SeamLines
 - Right Click & Reverse Sewing

M:N SEWING (Multiple to Numerous)

- Click and hold the tool icon and choose M:N
- Note the icon image changes, and text beside your cursor.
 - Indicates that you're in M:N Sewing.
 - Click and hold the tool icon to go back or use the hot keys (N) Segment Sewing or (M) Free Sewing.
- M:N Free Sewing on the armhole. Works like regular Free Sewing.
- Sew across sleeve cap in order, clicking to start and clicking to end.
 - NO need to hold SHIFT.
 - Press ENTER on my keyboard after I complete the sleeve cap.
 - Next move around the armhole, press enter after I complete all the armhole pieces.
 - Sew in the same order, and pay attention to Start marks.
- M:N Segment Sewing on the bottom band, sewing it to the body.
 - Works the same exact way
 - Click on each Segment and press enter when for the bottom band.
 - Same order, noting Start marks to the same side and press enter when I complete all the body pieces.

SEWING PROPERTIES

- Each SeamLine Pair Group has its own properties in the Property Editor.
- Sewing Line Type. Default all sewing is Custom Angle of 180 degrees; flat
 - All seams are effectively pressed open, like normal body seams on a garment.
 - If something is sewn to an internal line, like a pocket, CLO automatically assigns a Turned Sewing Line Type.
 - Turn the sewing in the Property Editor.
 - Neck trim and internal neck trim facing.
 - Top and bottom edges set to 180 angle because they are the edge of the fabric and CLO does not know to turn them.
 - Neck trim is puffy and rounded.
 - Change to Turned to flatten after simulation.
 - Most common; facing pieces, collar and undercollar, a lapel facing, a cuff and an under cuff.
- Tension. Default setting strength of .1 and ratio of 100. This is your seam tension.
 - Strength is the tightness of those stitches. Higher you go, the more tension
 - Turn the strength up, my seams start to hold tight and pull on my garment.
 - Uncheck Tension if it is affecting the drape of the garment.
 - Ratio default is 100; neither stretched nor eased when sewn.
 - Above 100%, the seams become stretched out. More intense; higher the strength
 - Less than 100% seams will shrink and be eased by the sewing. Much less significant strength down to .1,
 - Suggestion; keep your strength at .1 and your ratio at 100

- 3D Seamline
 - Normal map applied to all sewing to make it look more 3-dimensional.
 - "Shadow" on the seamline.
 - Change either the intensity or the thickness to 0 OR remove with trash icon to turn off
 - Thickness is the width of the normal map. Default is .059.
 - Increase this number, the line becomes thicker.
 - Most common use is default setting or turning it off by bringing one of the values down to 0.